Meridian Mart

Retail Sales, Profitability & Operations - Analysis Methodology

Platform: Tableau Desktop
Analysis period: 2022–2025
Scope: US retail; Central, East, South and West regions
Deliverables: Executive Performance dashboard; Profitability & Operations dashboard

1. Project Overview

I developed a Tableau reporting solution to assess Meridian Mart's retail sales, profitability and operational performance. Rather than relying on revenue as a standalone signal, I analysed profit margins, regional and product-level contributions, discount patterns, customer segments and shipment performance. I organised the output into two complementary dashboards: one for executive performance and one for deeper profitability and operational review.

The work uses the Meridian Mart business scenario and accompanying analytical dataset. Results are described as findings from that dataset, not as claims about an external live retail company.

2. Business Problem and Objectives

The core challenge was limited visibility across sales, products, geography and customer activity. Aggregated sales performance did not sufficiently explain which product lines or locations were generating profit, and inconsistent analysis made it harder to understand whether promotional and shipping decisions supported profitable growth.

The analytical objectives were to:

1. Measure total sales, profit, profit margin and distinct orders.
2. Compare performance by year, month, state, city, category, subcategory and customer segment.
3. Identify and quantify loss-making subcategories.
4. Compare discount bands with observed average profit and examine discount patterns by geography, category and segment.
5. Evaluate delivery times and shipping-mode order volumes.
6. Translate the results into decision-focused recommendations.

3. Dataset Description and Scope

The documented analytical model includes a central Sales table together with LocationDim, CustomerDim and ProductDim. These dimensions provide the geographic, customer and product classifications needed for the analysis. The source spans four displayed years (2022–2025) and four US regions.

Data area	Core fields / analytical use
Transactions	Order ID; Order Date; Ship Date; Ship Mode; Sales; Profit; Quantity; Discount
Customers	Customer ID; Customer Name; Segment
Products	Product ID; Category; Sub-Category; Product Name
Geography	Location ID; City; State; Region

Granularity: Sales and profit originate from sales records; `COUNTD([Order ID])` is used for unique orders rather than counting sales lines.

4. Data Preparation and Analytical Modelling

The analysis uses the documented structure of the Sales fact table related to customer, product and location dimensions through their identifier fields. This approach supports slicing sales and profit by segment, product and geographical attributes while retaining a transaction-centred set of metrics.

I used Tableau's native aggregations and calculated fields to create commercial and operational measures, including distinct order counts, profit margin, shipping-day calculations, average profit by discount band and subcategory profit. FIXED level-of-detail expressions enabled stable subgroup calculations, such as subcategory-level total profit, independently of individual visual mark granularity.

Validation controls relevant to the model

- I checked the uniqueness and compatibility of Customer ID, Product ID and Location ID across related tables.
- Compared raw sales totals with Tableau `SUM([Sales])` and profit totals with `SUM([Profit])` under equivalent filters.
- Verified that distinct order counts are computed using `COUNTD([Order ID])` and reconciled any view-level filtering.
- Inspected missing dates, discounts outside expected ranges, and shipment dates preceding order dates.
- Reviewed whether calculations using FIXED LOD need context filters to reflect the intended dashboard selection.

5. Tableau Measures and Calculation Logic

The principal measures inspected in the supplied Tableau workbook include:

Analytical measure	Tableau expression / logic	Purpose
Total Sales	`SUM([Sales])`	Total recorded revenue
Total Profit	`SUM([Profit])`	Total recorded profitability
Profit Margin	`SUM([Profit]) / SUM([Sales])`	Profit as a proportion of sales
Order Count	`COUNTD([Order ID])`	Distinct orders
Delivery Days	`DATEDIFF("day", [Order Date], [Ship Date])`	Days between ordering and shipping
Sub-category Profit	`{ FIXED [Sub-Category] : SUM([Profit]) }`	Stable subcategory profitability
Average Profit by Discount	`{ FIXED [Discount (bin)] : AVG([Profit]) }`	Compare discount bins
Profit per Order by Ship Mode	FIXED on Ship Mode; SUM(Profit) / COUNTD(Order ID)	Shipping profitability measure

The full field definitions and workbook-specific internal references are documented in the separate Meridian Mart Tableau Calculated Fields file.

6. Dashboard Design and Development

6.1 Executive Performance dashboard

The Executive Performance page presents four headline KPIs (Sales, Profit, Profit Margin and Order Count), a geographical view, monthly and yearly trends, subcategory performance, city-level profit comparisons and segment performance. Category and Year selectors support examination of different parts of the business.

This layout answers the executive questions: how much the business sold, how much it earned, which markets and categories contributed most, and whether revenue and profit were improving together.

6.2 Profitability & Operations dashboard

The second dashboard extends beyond headline performance into two loss-making subcategories, aggregate subcategory losses, average profit at observed discount levels, discount patterns by business grouping, delivery days by mode and region, ship-mode distribution, and top customers by sales and profit. Region and Segment selectors allow a focused operational review.

This page was designed to turn broad performance gaps into areas for investigation. In particular, negative profit across Tables and Machines is identified for follow-up, not presented as proof of a specific root cause.

7. Key Findings and Interpretation

7.1 Financial performance

Meridian Mart generated $4.47M in sales and $480.05K in profit, achieving a 10.74% profit margin across 2,100 orders. This indicates positive overall financial performance, although losses in certain products and locations suggest opportunities to improve profitability.7.2

Regional and city-level profitability

California recorded the highest sales ($531.8K) and profit ($66.6K), followed by New York. However, Colorado generated $119.4K in sales while recording a $4.5K loss. This shows that strong sales do not always translate into profitability, highlighting the need to assess pricing, discounts and costs at the regional level.

7.3 Category and subcategory performance

Copiers were a major revenue contributor, generating approximately $1.28M in sales. In contrast, Tables and Machines recorded combined losses of $125.2K, with Tables accounting for 70.9%. This identifies Tables as the greatest source of losses among the subcategories, making it a priority for investigating product costs, pricing and discount practices.

7.4 Discount and margin patterns

Average profit declined from $379 at 0% discount to $81 at 20%, before turning negative at 40%. This suggests that higher discount levels are associated with declining profitability. Discount controls may therefore help protect margins, although further analysis is needed to establish profitable thresholds for individual products.

7.5 Customer segments

The Consumer segment generated the highest revenue ($2.22M) and profit ($221.1K). However, Home Office achieved the highest estimated profit margin at 12.4%, despite having the lowest sales. This indicates that Consumer drives overall profitability through volume, while Home Office offers potential for higher-margin growth.

7.6 Time trends

Meridian Mart recorded its strongest performance in 2024, generating $1.22M in sales and $151.5K in profit. Sales subsequently declined by approximately 17% in 2025. This signals weakening sales performance following the company's strongest year, highlighting the need to investigate demand changes and identify opportunities for recovery without sacrificing margins.

7.7 Fulfilment and customer concentration

Standard Class handled the highest order volume (1,235 orders) but recorded the longest average delivery time of 5.4–5.5 days. First Class generated the highest reported profit per order ($301), while Same Day offered the fastest delivery at 0.5–0.6 days. This shows

differences in shipping efficiency and profitability, suggesting opportunities to review delivery costs, service pricing and customer preferences.

8. Strategic Recommendations

1. Prioritise Tables and Machines for deeper investigation. Review individual products, actual costs, selling prices, shipping expense and discounts before changing assortment or pricing.
2. Introduce evidence-based discount monitoring. Flag deep discounts for additional review, compare similar products, and use actual contribution margins to assess approvals rather than relying only on the displayed threshold.
3. Investigate Colorado and Denver losses. Break down the loss by product mix, customer segment and transaction to identify whether profitability can improve without weakening overall sales.
4. Preserve successful demand and margin combinations. Investigate the product and customer strategies behind stronger-performing regions and 2024 results before adapting them elsewhere.
5. Develop customer segment actions. Support the Consumer segment's large contribution while testing high-margin opportunities in Home Office without unnecessary discounting.
6. Strengthen fulfilment reviews. Track delivery speed, order volumes, exceptions and actual distribution costs for Standard and premium shipping classes.
7. Make margin a core executive KPI. Review profit margin regularly alongside revenue by city, state, category and segment.

9. Conclusion

The Meridian Mart analysis demonstrates how Tableau can combine financial, customer, product and fulfilment measures into an accessible decision-support system. The dashboards reveal that strong overall revenue can coexist with geographic losses, subcategory-level erosion and discount-linked margin pressure. By examining these patterns together, the analysis identifies clearer priorities for pricing reviews, product evaluation, customer strategy and shipping performance.

The main value of the project is not simply presenting KPIs, but using them to identify where commercial decisions warrant further testing and action.